# SOFTWARE CHANGE IMPLEMENTATION REPORT

## CR-033: Last Login Tracking and Display for Student Accounts

| Field | Details |
|---|---|
| Change Request ID | CR-033 |
| Change Request Title | Last Login Tracking and Display for Student Accounts |
| Student Name | T.Manoja |
| Student ID | JF/ICT/24/12 |
| Implementation Status | Completed |
| Date | 13 October 2026 |

# 1. Change Request Summary

## 1.1 Business Requirement

The Student Management System required a facility to track and display the most recent successful login time of student accounts. This information helps administrators identify student account activity and understand when students last accessed the system.

Before this change, the system stored student account details and authentication information but did not record or display the last successful login time. Therefore, administrators could not easily identify whether a student had logged into the system recently.

The change was required to record the last successful login time for student accounts and display that information in the administrator's student list without affecting existing authentication or student management functionality.

## 1.2 Existing Behaviour

- Student account information was stored in the database.
- Students could log in using their registered credentials.
- The authentication system verified usernames and passwords.
- The administrator could view student details through the Admin Dashboard.
- The system did not record the last successful login time for student accounts.
- The Admin Dashboard did not contain a Last Login column.
- Administrators could not distinguish between students who had logged in recently and students who had never logged in.

## 1.3 Requested Change

- Add a nullable last_login field to the users table.
- Update the timestamp only after successful student authentication.
- Do not update the timestamp for incorrect passwords or nonexistent usernames.
- Return last_login through existing student retrieval functionality.
- Display a Last Login column in the Admin Dashboard.
- Display “Never logged in” when the timestamp is NULL.
- Preserve existing authentication, token generation, student search, and CRUD functionality.

# 2. Implementation

## 2.1 Implementation Summary

CR-033 was implemented by extending the existing Student Management System backend and frontend. A new last_login column was added to the users database table to store the most recent successful login date and time. The column allows a NULL value for accounts that have not yet recorded a successful login.

The user model was updated with an updateLastLogin() function. The authentication controller calls this function after the submitted username and password have been verified. Failed authentication attempts do not trigger the timestamp update.

The student model retrieves last_login by joining the students and users tables. The existing student retrieval endpoint returns the timestamp alongside student details. The Admin Dashboard displays a Last Login column and formats valid timestamps as readable date and time. If the timestamp is missing or NULL, it displays “Never logged in”. No separate login-tracking API endpoint is required.

## 2.2 Affected Files

| File / Location | Change | Purpose |
|---|---|---|
| database/schema.sql | Modified | Adds the last_login column for new database installations. |
| backend/src/models/userModel.js | Modified | Adds updateLastLogin() and exports it with existing model functions. |
| backend/src/controllers/authController.js | Modified | Updates the timestamp after successful student authentication. |
| backend/src/models/studentModel.js | Modified | Retrieves last_login by joining student records with user accounts. |
| frontend/src/pages/DashboardPage.jsx | Modified | Adds the Last Login column and displays formatted timestamps or “Never logged in”. |

## 2.3 Implementation Logic

1. Add the last_login column to the users table.
2. Retain the existing username and password authentication process.
3. Retrieve the user account using the submitted username.
4. Verify the submitted password against the stored password hash.
5. If authentication fails, return the existing error response without updating the timestamp.
6. If a student authenticates successfully, update last_login using the current database date and time.
7. Continue with existing JWT generation and the successful login response.
8. Retrieve student information together with the associated user's last_login field.
9. Return student information through the existing student retrieval endpoint.
10. Display the Last Login column in the Admin Dashboard.
11. Format a valid timestamp into readable date and time.
12. Display “Never logged in” when the timestamp is NULL.
13. Verify existing student management and authentication functionality.

## 2.4 Error and Data Handling

- Successful student login: update the student's last login timestamp.
- Failed student login: do not update the timestamp.
- No recorded login: display “Never logged in”.
- Existing student information: preserve existing fields and records.
- Existing CRUD operations: retain create, read, update, and delete functionality.
- Administrator authentication: retain the existing process without applying the student-specific update.

- Missing or invalid timestamp: avoid displaying an invalid formatted date.

The change requires a database schema update and backend code changes; it is not a frontend-only enhancement.

# 3. Testing & Verification

## 3.1 Testing Approach

Verification covers the database, backend authentication, student information retrieval, and Admin Dashboard. Testing should confirm successful student logins update the timestamp, failed logins do not, the timestamp displays correctly, and existing functionality remains operational.

Testing types: Functional, database, authentication, API response, UI, data retrieval, error-handling, and regression testing.

Environment: Development | Browser: Chrome / Microsoft Edge | Backend: Node.js and Express.js | Database: MySQL | Test roles: Administrator and Student

## 3.2 Test Cases

| Test ID | Test Case | Expected Result | Actual Result | Status |
|---|---|---|---|---|
| TC01 | Verify last_login column | Users table contains the new nullable timestamp field. | Column verified in database. | Pass |
| TC02 | Successful student login | Student's last_login value is updated. | Timestamp updated after successful student login. | Pass |
| TC03 | Incorrect password | Login fails and timestamp remains unchanged. | Login rejected; timestamp unchanged. | Pass |
| TC04 | Nonexistent username | Login fails without updating any student timestamp. | Login rejected without timestamp update. | Pass |
| TC05 | Second successful login | Timestamp updates to latest successful login time. | Timestamp updated to latest login time. | Pass |
| TC06 | Retrieve administrator's student list | Student information includes last_login. | Student list included last_login. | Pass |
| TC07 | Open Admin Dashboard | Last Login column appears. | Column displayed correctly. | Pass |
| TC08 | NULL timestamp | Dashboard displays “Never logged in”. | “Never logged in” displayed. | Pass |
| TC09 | Valid timestamp | Dashboard displays readable date and time. | Timestamp displayed in readable format. | Pass |
| TC10 | Navigate and search student | Listing, navigation, and search remain | Listing, navigation, and search worked. | Pass |

| | | | | |
|---|---|---|---|---|
| | list | operational. | | |
| TC11 | Edit and Delete student | Existing CRUD functionality continues. | Edit and Delete continued to work. | Pass |
| TC12 | Login response and token | Successful authentication returns expected response and token. | Expected response and token returned. | Pass |

### 3.3 Test Summary

| Test Result | Count |
|---|---|
| Total Test Cases | 12 |
| Passed | 12 |
| Failed | 0 |
| Pass Rate | 100% |

Overall Test Result: Pass

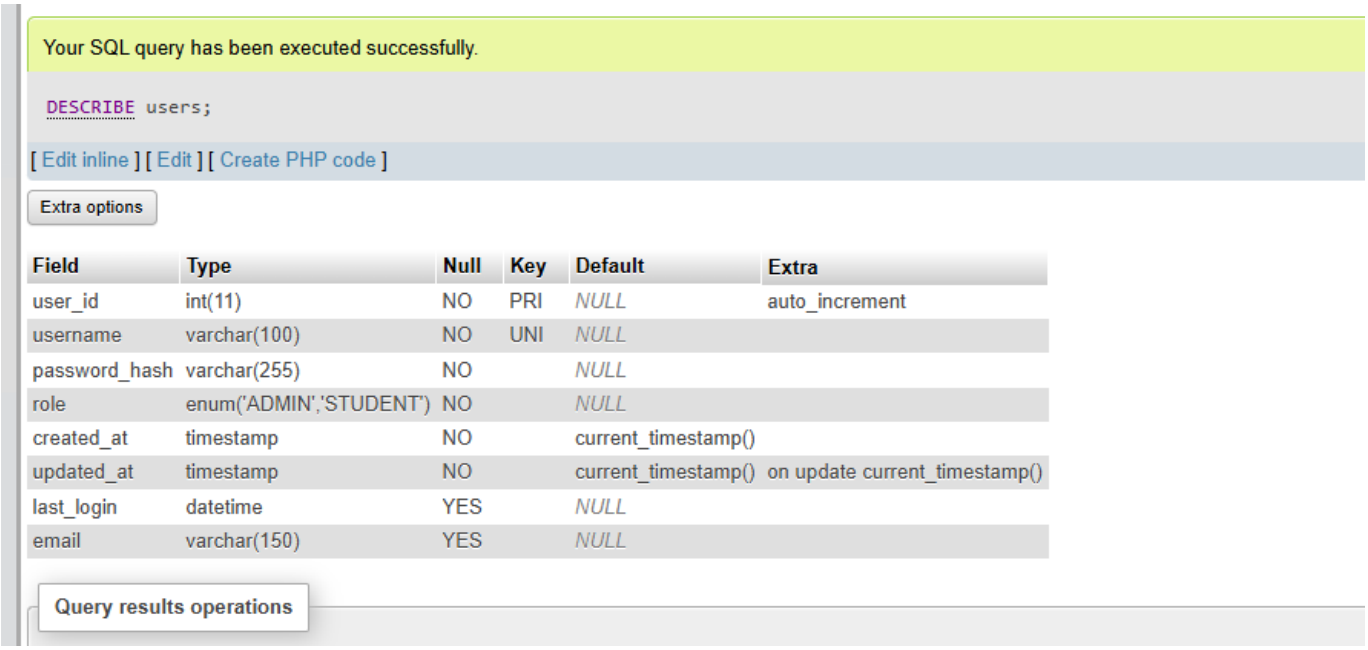
## 4. Acceptance Criteria Verification

| ID | Acceptance Criterion | Verification Method | Status |
|---|---|---|---|
| AC001 | Database stores last successful student login timestamp. | Database query | Met |
| AC002 | Successful student login updates the timestamp. | Login and database verification | Met |
| AC003 | Failed login does not update the timestamp. | Failed login and database comparison | Met |
| AC004 | Student retrieval includes last_login. | API response inspection | Met |
| AC005 | Admin Dashboard displays Last Login column. | UI inspection | Met |
| AC006 | Accounts without login display “Never logged in”. | UI and database verification | Met |
| AC007 | Timestamp is readable and correctly formatted. | UI inspection | Met |
| AC008 | Authentication and token generation remain operational. | Regression testing | Met |
| AC009 | Existing student CRUD remains operational. | CRUD regression testing | Met |

# 5. Evidence

### Evidence 01 – Database Last Login Column

Related Acceptance Criteria: AC-001



Your SQL query has been executed successfully.

```
DESCRIBE users;
```

[ Edit inline ] [ Edit ] [ Create PHP code ]

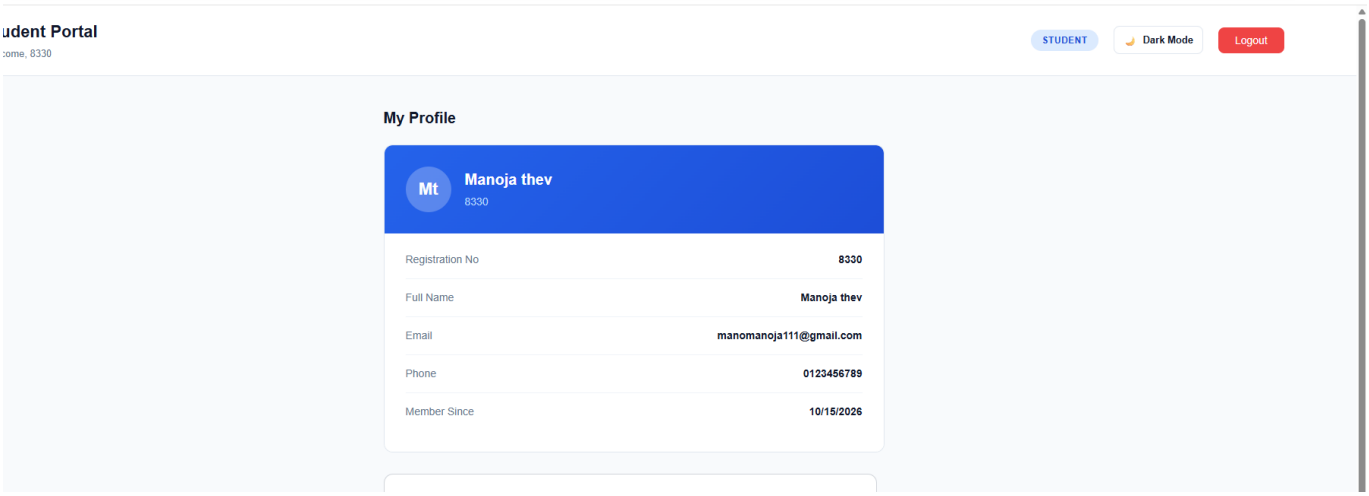
Extra options

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| user_id | int(11) | NO | PRI | *NULL* | auto_increment |
| username | varchar(100) | NO | UNI | *NULL* | |
| password_hash | varchar(255) | NO | | *NULL* | |
| role | enum('ADMIN','STUDENT') | NO | | *NULL* | |
| created_at | timestamp | NO | | current_timestamp() | |
| updated_at | timestamp | NO | | current_timestamp() | on update current_timestamp() |
| last_login | datetime | YES | | *NULL* | |
| email | varchar(150) | YES | | *NULL* | |

Query results operations

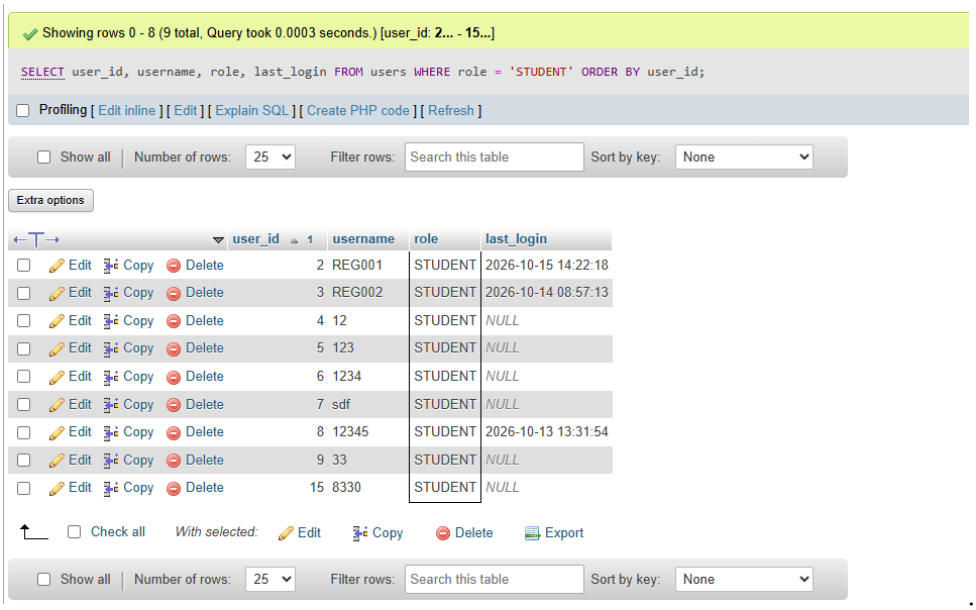
### Evidence 02 – Successful Student Login

Related Acceptance Criteria: AC-002, AC-008

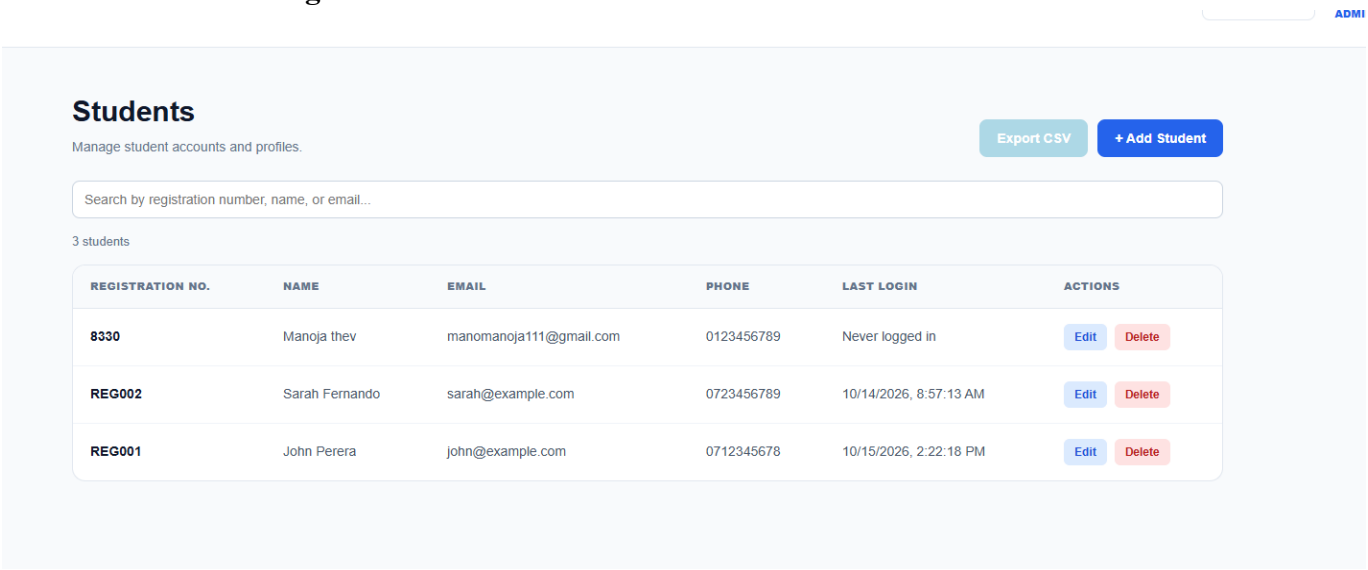



### Evidence 03 – Failed Login Does Not Update Timestamp

Related Acceptance Criteria: AC-003

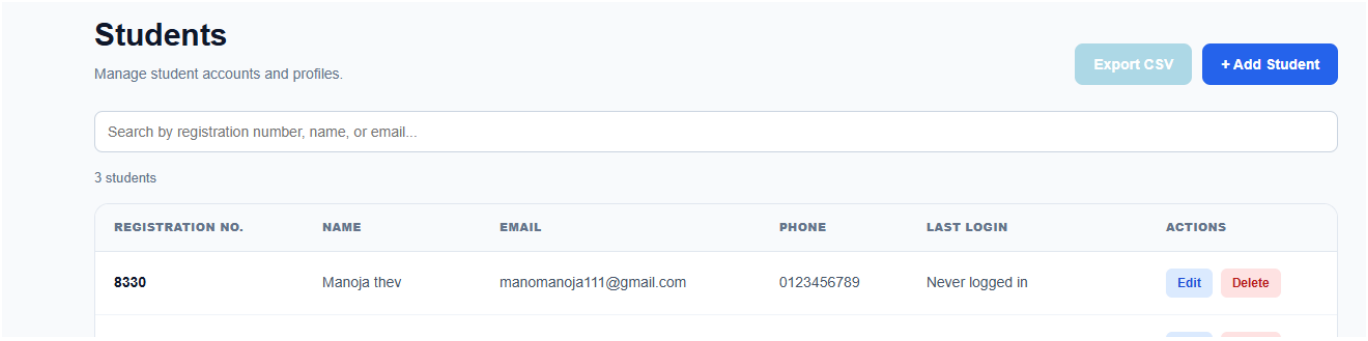



**Evidence 04 – Last Login Column in Admin Dashboard**




**Evidence 05 – Never Logged In State**

Related Acceptance Criteria: AC-006




# 6. Final Result

| Item | Result |
|---|---|

| | |
|---|---|
| Requirements | Met |
| Implementation | Completed |
| Testing | Passed |
| Acceptance Criteria | All Met |
| Existing Functionality | Verified |
| Final Status | Accepted |

## 6.1 Final Summary

CR-033 introduces last login tracking and display functionality for student accounts in the Student Management System. The last_login field is added to the users table and updated after successful student authentication. The controller performs the update after password verification, while failed login attempts do not update the timestamp.

The student model retrieves the timestamp through a join between students and users. The Admin Dashboard displays a dedicated Last Login column. Accounts with a recorded login show readable date and time, while accounts without a timestamp display “Never logged in”.

The change is designed to preserve existing login, token generation, student retrieval, search, Edit, and Delete functionality. It requires a database migration and modifications to existing backend and frontend files.

Final completion and acceptance should be recorded after the database migration, successful and failed login scenarios, dashboard display, and regression tests have actually been executed and their results documented.

## 6.2 Issues / Limitations

**The last_login column must exist before updated backend code can use it.**

14. Existing accounts show “Never logged in” until a successful student login is recorded.
15. The feature records only the most recent successful student login; it does not maintain full login history.
16. The displayed time uses browser date/time formatting. Consider the database server time zone when checking exact values.
17. The current implementation updates student accounts only. Administrator login tracking is outside this change request.
18. Test results and final acceptance status must be based on actual execution in the development environment.

**Student Name:** T.Manoja

**Student ID:** JF/ICT/24/12

**Date:** 13 October 2026

**Signature**